

Numbers and Sets

ECON 441: Introduction to Mathematical Economics

Lecture 1

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Introductions

- Your preferred name and pronouns
- Why are you here?
 - Undergrads: why you signed up for this class
 - Grad students: why you're doing the MA
- The last math class you took, and roughly when
- Your comfort food

Real-Number System

- *Integers*:

$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$

- *Fractions*:

$\frac{1}{2}, \frac{3}{5}, -\frac{2}{3}$

- *Rational numbers*: ratio of integers
- *Are fractions rational numbers? What about integers?*

Real-Number System

- *Rational numbers*: ratio of integers
 - “terminating or repeating decimal”
 - Example. $\frac{1}{3} = 0.333$, $\frac{1}{4} = 0.25$
- *Irrational numbers*: cannot be expressed as a ratio of two integers
 - “nonterminating or nonrepeating decimal”
 - Example. $\sqrt{2} = 1.4142$, $\pi = 3.1415$
- *Real numbers* ($\mathbb{R}$): rational and irrational

Sets

- A *set* is a collection of distinct objects.

$$A = \{brownies, icecream, pizza, ramen\}$$

- $pizza \in A$, $\in$ stands for ‘is in’
- What about sets B and C ?

$$B = \{x | x \text{ is a positive integer}\}$$

$$C = \{x | 1 < x < 5\}$$

Set Relations

- Equivalence ($=$)

$$A = \{brownies, icecream, pizza, ramen\}$$

$$B = \{pizza, ramen, icecream, brownies\}$$

$$A = B$$

- Subset ($\subset$)

$$C = \{pizza, ramen\}$$

$C \neq A$ but $C \subset A$. Note: $A \supset C$ is equivalent. Is $A \subset B$? Yes, but C is a proper subset of A .

Set Relations

- Disjoint sets

$$A = \{brownies, icecream, pizza, ramen\}$$

$$D = \{salad, fruits\}$$

- Neither but still related

$$A = \{brownies, icecream, pizza, ramen\}$$

$$E = \{salad, fruits, icecream\}$$

Set Relations

- $\emptyset$: empty or null set
- What are all possible subsets of $S = \{a, b, c\}$?
 - $\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}$
 - Always 2^n subsets. Here $n = 3$, so 8 subsets.

Set Operations

- *Union:* $A \cup B$, elements in either A or B
- *Intersection:* $A \cap B$, elements in both A and B

Example:

$$A = \{\text{brownies}, \text{icecream}, \text{pizza}, \text{ramen}\}$$

$$B = \{\text{salad}, \text{fruits}, \text{icecream}\}$$

$$A \cup B = A \cap B =$$

Set Operations

- *Union:* $A \cup B$, elements in either A or B
- *Intersection:* $A \cap B$, elements in both A and B

What about

$$A = \{\text{brownies}, \text{icecream}, \text{pizza}, \text{ramen}\}$$

$$B = \{\text{salad}, \text{fruits}\}$$

$$A \cup B = A \cap B =$$

Set Operations

- *Complement of A:* $\tilde{A}$ or A^c , 'not A '
- Universal set U (context specific) then:

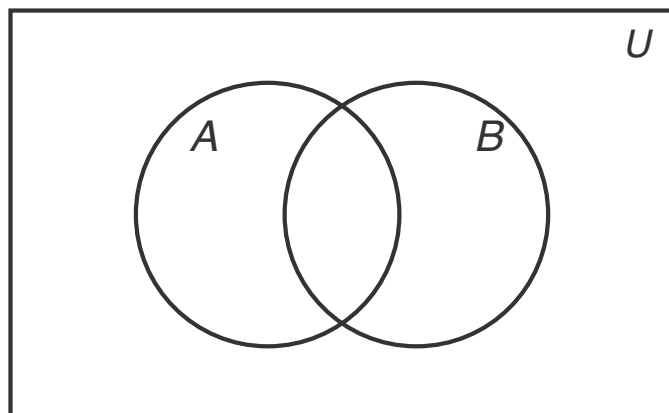
$$A^c = \{x | x \in U \text{ and } x \notin A\}$$

Example. $U = \{1, e, f, 2\}$, $A = \{1, 2\}$, then $A^c = \{e, f\}$.

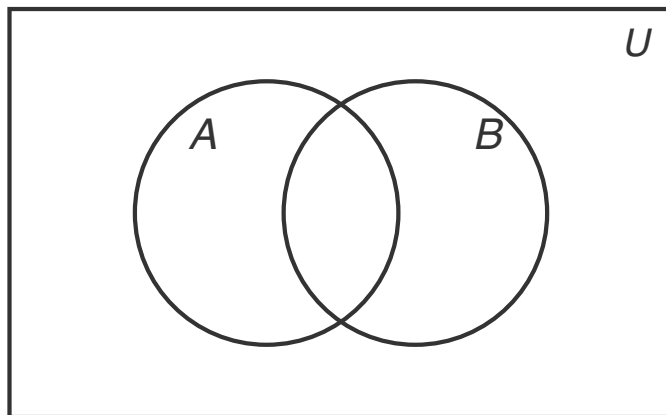
Set Operations: Venn Diagrams

Shade the following regions:

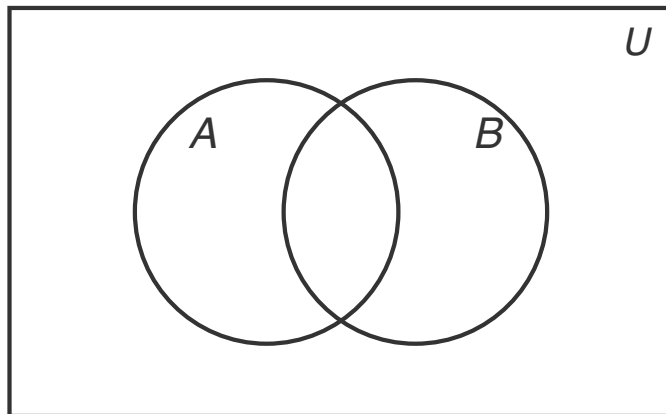
$$A \cup B$$



$A \cap B$



A^c



Laws of Set Operations

- Commutative law

$$A \cup B = B \cup A \quad A \cap B = B \cap A$$

- Distributive law

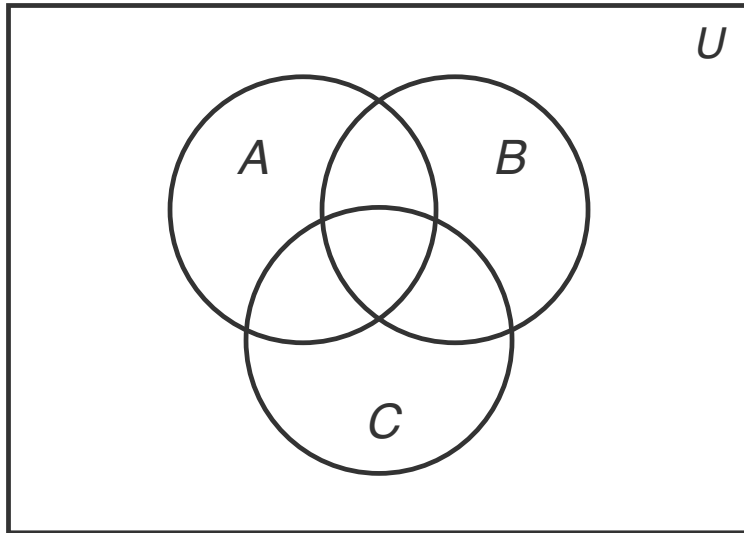
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

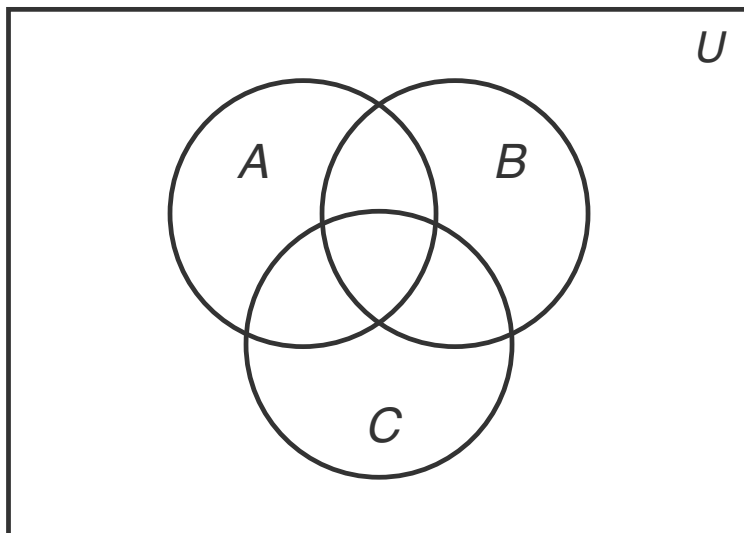
Distributive law

Verify by shading: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

A



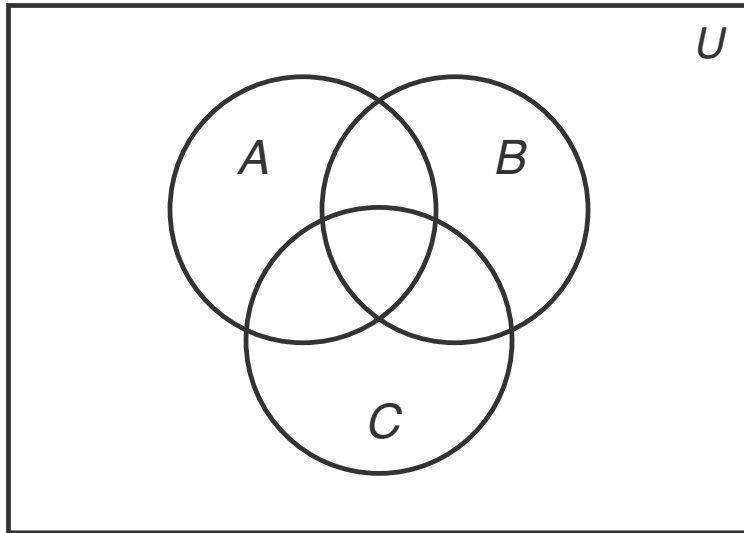
$B \cap C$



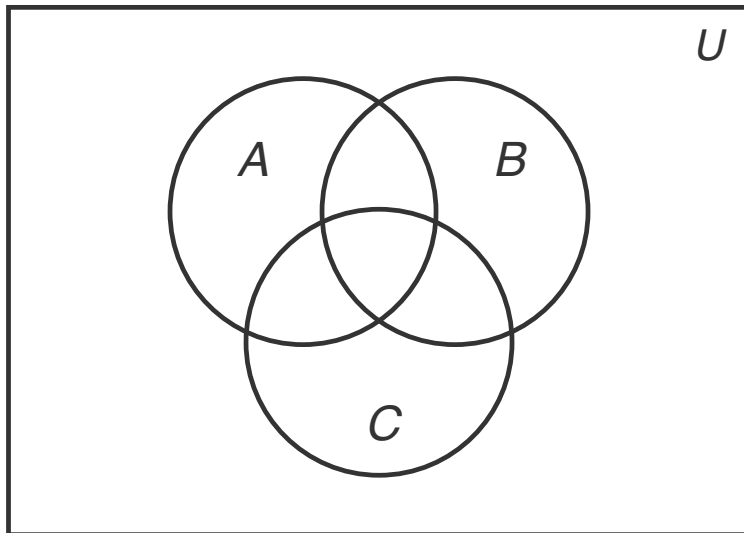
Distributive law

Now the right-hand side, $(A \cup B) \cap (A \cup C)$:

$A \cup B$



$A \cup C$



Ordered Sets

- We said order does not matter for sets
- But we can have ordered sets where

$$(a, b) \neq (b, a) \text{ unless } a = b$$

- Ordered pairs, triples,...

Example. $(age, weight)$, $(22, 120)$ different from $(120, 22)$

Cartesian Product

$$A = \{1, 2\} \quad B = \{3, 4\}$$

Cartesian Product: set of all possible ordered pairs

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

Cartesian Plane

$$\mathbb{R}^2 = \{(x, y) | x \in \mathbb{R}, y \in \mathbb{R}\}$$

Can have $\mathbb{R}^3, \mathbb{R}^4, \dots, \mathbb{R}^n$

By the way

What is $x \times x$? What is $x^2 \times x$? What is $x^2 \times x^2$? More generally, $x^n \times x^m = x^{m+n}$